

Targeted therapies for soft tissue sarcoma

Targeted therapy drugs interfere with the way that cancer cells grow. They are often given as part of research trials and are sometimes used as a treatment for soft tissue sarcomas.

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About targeted therapy for soft tissue sarcoma

[Targeted therapy](#) uses drugs to find and attack cancer cells. There are many different types of targeted therapy, including tyrosine kinase inhibitors (TKIs). Each type targets something in or around the cancer cell that is helping it grow and survive. They may be used to help to treat gastrointestinal stromal tumours gastrointestinal stromal tumour (GIST) or to control [soft tissue sarcoma](#) that has spread.

Targeted therapy drugs may be given as part of a [research trial](#) to treat soft tissue sarcoma.

Imatinib (Glivec®)

[Imatinib](#) is a TKI. It may be used to treat [GISTs](#), or a rare type of soft tissue sarcoma called dermatofibrosarcoma protuberans. Imatinib works by blocking (inhibiting) signals inside cancer cells. Blocking the signals stops a series of chemical reactions that make the cells grow and divide. The chemical it blocks is called tyrosine kinase.

It may be given [before surgery](#) to shrink a GIST and make surgery more effective. If there is a higher risk of GIST coming back you usually have imatinib for 3 years. This is called adjuvant therapy.

Imatinib may also be given if you have a GIST that cannot be operated on or has spread to other parts of the body. It may help to control it for several years.

You take imatinib once a day as tablets. It can cause side effects including:

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[tiredness](#) (fatigue)

- [feeling or being sick](#)
- [diarrhoea](#)
- a skin rash
- puffiness, especially around the eyes.

Sunitinib (Sutent®)

[Sunitinib](#) is another TKI. It may be used to treat GISTs that do not respond to imatinib, or if you get a lot of side effects with imatinib.

Sunitinib blocks signals in the cancer cells. This stops a series of chemical reactions that make the cell grow and divide. It also helps stop the tumour developing new blood vessels. Without blood vessels, the tumour cannot get the nutrients it needs to survive.

You take sunitinib as a tablet. Some possible side effects include:

- [tiredness](#) (fatigue)
- high blood pressure
- [diarrhoea](#)
- sore hands and feet
- a skin rash
- changes to your hair colour
- thyroid changes.

Regorafenib (Stivarga®)

[Regorafenib](#) is a newer TKI. It may sometimes be used after imatinib and sunitinib to treat an advanced GIST that cannot be removed with surgery. It may shrink the tumour or stop it growing for a time. Regorafenib is taken as tablets.

Some possible side effects include:

- an increased risk of infection
- [loss of appetite](#)
- [diarrhoea](#)
- [feeling or being sick](#)
- a rash
- sore hands and feet
- hoarse voice

- high blood pressure.

Other targeted therapy drugs

Pazopanib (Votrient®)

[Pazopanib](#) is another TKI. It may be used to treat some types of advanced soft tissue sarcoma. It is taken once a day as tablets.

Possible side effects include:

- high blood pressure
- [diarrhoea](#)
- [feeling or being sick](#)
- [tiredness](#) (fatigue)
- a skin rash
- hair colour changes.

Pazopanib has not been approved by the [National Institute for Health and Care Excellence \(NICE\)](#) or the [Scottish Medicines Consortium \(SMC\)](#) for use on the NHS. Because of this, it may not be widely available.

Crizotinib (Xalkori®)

[Crizotinib](#) may be used to treat a rare type of sarcoma called a myofibroblastic tumour. It is not available on the NHS but may be used in certain circumstances. You take it as capsules at home.

Larotrectinib (Vitrakvi®)

[Larotrectinib](#) can be used to treat cancers that have a change in a gene called NTRK. But this is not common in sarcoma, so it is rarely used. You have tests on the cancer cells to check for the gene change. You take it as capsules or sometimes as a liquid.

More information

We have more general information about [targeted therapy](#). On this page you can also watch a video about how targeted therapy works.

About our information

References



Below is a sample of the sources used in our soft tissue sarcoma information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Gronchi A, Miah AB et al. Soft tissue and visceral sarcomas: ESMO-EURACAN-GENTURIS Clinical practice guidelines for diagnosis, treatment and follow-up. *Annals of Oncology*, 2021; 32, 11, 1348-1365 [accessed May 2022].

Casali PG, Blay JY et al. Gastrointestinal stromal tumours: ESMO-EURACAN-GENTURIS Clinical practice guidelines for diagnosis, treatment and follow-up. *Annals of Oncology*, 2022; 33,1, 20-33 [accessed May 2022].

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by senior medical editor [Fiona Cowie](#), Consultant Clinical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

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ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

[Get help](#)